

MATH120 Discrete Mathematics

Tutorial 7

References

- Topics 11-12: Sets and Relations
- Sections 1.2, 6.1–3, 8.1–3 of Epp (4th or 5th Ed.)

Work on the following problems during the tutorial in week 8.

1. [B] Consider the sets

$$\begin{aligned}A &= \{n \in \mathbb{Z}^+ : n \text{ is odd}\} \\B &= \{n \in \mathbb{Z}^+ : n \text{ is prime}\} \\C &= \{4n + 3 : n \in \mathbb{Z}^+\}\end{aligned}$$

Which of these sets are subsets of which? Consider all 9 possibilities (consider 4 possibilities in class and the remainder as homework).

$\Rightarrow$

2. [M] Each of the following statements about sets A , B , and C are false. Give a counterexample to each statement.

- (a) If $A \cap B = A \cap C$, then $B = C$.
- (b) If $A \subseteq B \cup C$, then either $A \subseteq B$ or $A \subseteq C$.

$\Rightarrow$

3. [M] Let A , B , and C be subsets of a universal set U .

- (a) Show that if $A \subseteq C$ and $B \subseteq C$, then $A \cup B \subseteq C$.
- (b) Show that $(A - B) \cap (C - B) = (A \cap C) - B$.

$\Rightarrow$

4. [B] Let $\mathcal{D}$ be the relation on $\mathbb{N}$ that is defined by setting

$$\mathcal{D} = \{(a, b) \in \mathbb{N} \times \mathbb{N} : b = an \text{ for some } n \in \mathbb{N}\}.$$

- (a) List five ordered pairs in $\mathcal{D}$.
- (b) List three ordered pairs in $\mathbb{N} \times \mathbb{N}$ that are **not** in $\mathcal{D}$.
- (c) Is $\mathcal{D}$ reflexive? Symmetric? Antisymmetric? Transitive?

$\Rightarrow$

5. [M] Let $\mathcal{R}_1$ and $\mathcal{R}_2$ be relations on a set A .
- (a) Show that if $\mathcal{R}_1$ and $\mathcal{R}_2$ are both symmetric, then the relation $\mathcal{R}_1 \cap \mathcal{R}_2$ on A is symmetric.
 - (b) Give an example of relations $\mathcal{R}_1$ and $\mathcal{R}_2$ such that $\mathcal{R}_1 \cup \mathcal{R}_2$ is symmetric but $\mathcal{R}_1$ is not symmetric (you may choose the set A).
6. [B-M] Let $U = \{a, b, d, c, d, e, f, g\}$, $A = \{a, d, e, f\}$, and $B = \{b, e, g\}$ be sets, where U is the universal set. Determine the following sets and for each set X , also determine $|X|$.

- | | | |
|--------------------|-------------|----------------------|
| (a) $(A \cup B)^c$ | (c) $A - B$ | (e) $\mathcal{P}(B)$ |
| (b) $A \cap B$ | (d) $B - A$ | |

7. [D-B] Let $A = \{a, b, c\}$ and $B = \{a, b, d\}$.
- (a) List the ordered pairs in $A \times B$ and determine $|A \times B|$.
 - (b) List the elements in the set $S = \{(x, y) \in A \times B : x = y\}$. Determine $|S|$.
8. [C] Prove or disprove the following statements. Assume all sets are subsets of a universal set U .
- (a) For all sets A , B , and C , if $A \subseteq B$ then $A \cap (B \cap C)^c = \emptyset$.
 - (b) For all sets A , B and C , if $A \cap C = B \cap C$ and $A \cup C = B \cup C$, then $A = B$.

$\Rightarrow$

9. [B] Let $\mathcal{D}$ be the relation on $\mathbb{Z}$ defined by $(m, n) \in \mathcal{D} \iff m + 2n \equiv 0 \pmod{3}$.
- (a) Show that $\mathcal{D}$ is an equivalence relation (i.e., that $\mathcal{D}$ is reflexive, symmetric and transitive).
 - (b) Determine the equivalence class $[1]$ determined by 1.
 - (c) What is the partition induced by $\mathcal{D}$ on $\mathbb{Z}$?
10. [B] Let $A = \{1, 2, 3, 4\}$. Give an example of a relation $\mathcal{R}$ on A that is
- (a) reflexive and symmetric, but not transitive,
 - (b) reflexive and transitive, but not symmetric,
 - (c) symmetric and transitive, but not reflexive.
11. [M] Let $\mathcal{R}_1$ and $\mathcal{R}_2$ be relations on a set A .

- (a) Suppose $\mathcal{R}_1$ and $\mathcal{R}_2$ are both reflexive. Show that $\mathcal{R}_1 \cup \mathcal{R}_2$ is always reflexive.
- (b) Give an example of relations $\mathcal{R}_1$ and $\mathcal{R}_2$ such that $\mathcal{R}_1 \cup \mathcal{R}_2$ is reflexive but $\mathcal{R}_1$ is not reflexive (you may choose the set A).

12. [B] Let $A = \{1, 2, 3, 4, 5, 6\}$ and let

$$\mathcal{R} = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4), (4, 5), (5, 4), (5, 5), (6, 6)\}$$

be an equivalence relation on A .

- (a) What are $[1]$, $[2]$, and $[3]$, the equivalence classes of 1, 2, and 3, respectively?
 - (b) What partition of A does $\mathcal{R}$ induce?
13. [C] Let $\mathcal{R}$ be an equivalence relation on a set A , and let $a, x, y \in A$. Prove that if $x, y \in [a]$, then $(x, y) \in \mathcal{R}$.